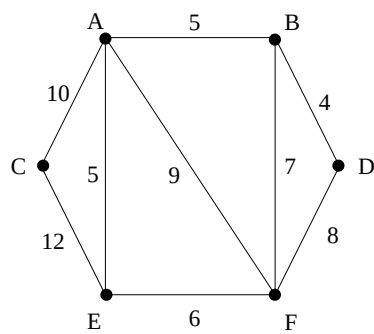


Discrete Mathematics: Network Algorithms revision

1 Show how to find a minimum spanning tree of the network shown below using:

(a) Prim’s algorithm, starting at A.

(b) Kruskal’s algorithm



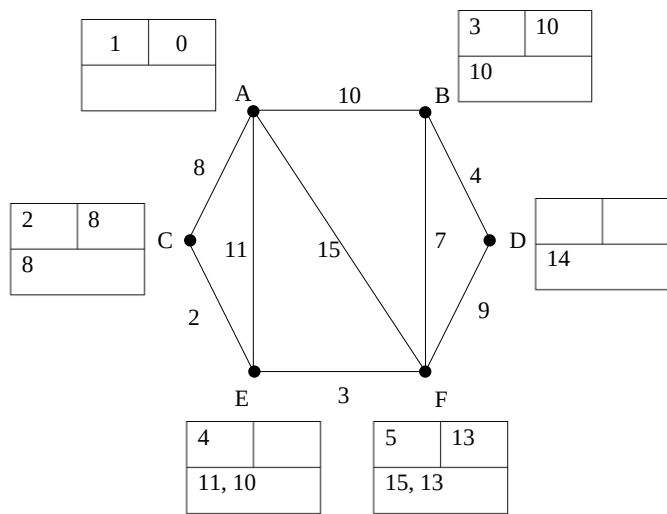
Commented [NS(1): Start A
Add B (AB=5)
Add D (BD=4)
Add E (AE=5)
Add F (EF=6)
Add C (AC=10)

Stop (six vertices)

Note that E could be added before B.

Commented [NS(2): BD=4
AB=5
AE=5
EF=6
BF=7 no (A-B-F-E-A cycle)
DF=8 no (A-B-D-F-E-A cycle)
AF=9 no (A-E-F-A cycle)
AC=10
STOP (5 edges)

2 Use Dijkstra’s algorithm to find the shortest path from A to F in the network shown below



Note the order labels of B and E can be interchanged, and D should not be made permanent.

- 3 Show how to use the tabular version of Prim's algorithm to find the minimum spanning tree of the network described the distance matrix below.

	A	B	C	D	E	F
A	-	8	10	-	12	5
B	8	-	6	18	2	-
C	10	6	-	7	-	9
D	-	18	7	-	6	-
E	12	2	-	6	-	3
F	5	-	9	-	3	-

- 4 Duke has written a computer program to implement the tabular version of Prim's algorithm. When he inputs a 50×50 distance matrix it takes his programme 78 milliseconds to output the minimum spanning tree. Estimate how long his program would take if he inputted a 200×200 distance matrix.

- 5 Given a network on n vertices, consider repeatedly using Dijkstra's algorithm to find the shortest distance between every pair of vertices. If it takes 2.4 seconds for a computer to do this for a complete network on 35 vertices, then roughly how long will it take the same computer do to this for a complete network on 100 vertices?

- 6 The weights on the edges of a complete network are sorted into ascending order using bubble sort. It takes a computer 182 milliseconds to do this for a complete network on 60 vertices. Roughly how long will it take the same computer to do this for a complete network on 500 vertices?

Commented [NS(3):

	A	B	C	D	E	F
A	-	8	10	-	12	5
B	8	-	6	18	2	-
C	10	6	-	7	-	9
D	-	18	7	-	6	-
E	12	2	-	6	-	3
F	5	-	9	-	3	-

Note that C and D could be selected in a different order.

Commented [NS(4): Prim's algorithm has order n^3 (you're expected to know this) so it will take approximately $78 * 200^3 / 50^3 = 4992$ milliseconds (~5 seconds)

Commented [NS(5): Dijkstra is order n^2 and we need to apply Dijkstra order n times, so this process will be order n^3 .

Therefore it should take roughly 56 seconds for a complete network on 100 vertices. So about 1 minute.

Commented [NS(6): There are $\frac{1}{2} n(n-1)$ edges in a complete network, that is, order n^2 edges. Bubble sort has quadratic order on the length of the list. Therefore this process has order n^4 .

Therefore it should take roughly fifteen minutes to complete the operation for a complete network on 500 vertices..